Tutorial – 05

Department of Physics **Shiv Nadar University**

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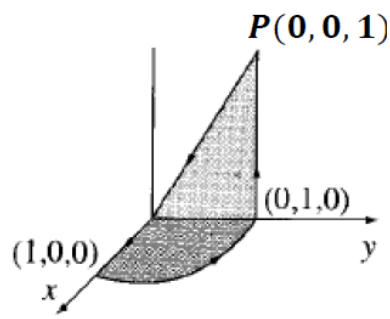
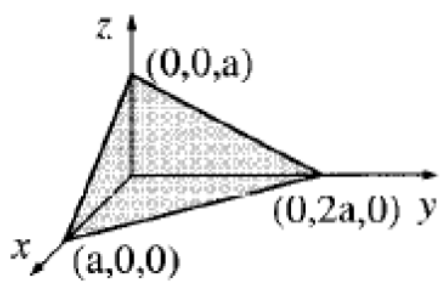
1. Verify Stoke's theorem for the vector function $\vec{\boldsymbol{v}}$ expressed in spherical co-ordinates (r, θ, φ) as:

$$\vec{\boldsymbol{v}} = (r cos^2\theta)\hat{\boldsymbol{r}} - (r\ cos\theta\ sin\theta)\hat{\boldsymbol{\theta}} + 3r\hat{\boldsymbol{\varphi}}$$

in the shaded region as shown



2. Check Stoke's theorem for the function $\vec{\boldsymbol{v}} = y\hat{\boldsymbol{z}}$ using the triangular surface as shown



3. Evaluate the following integrals:

(a) $\int_{-\infty}^{+\infty} e^{-x}\delta(x - ln2)dx$

(b) $\int_{1}^{3} x^3\delta(x - \pi)dx$

(c) $\int_{-1/2}^{7/2} \sin x\ \ \delta(x - \pi)dx$